



Grade 10 Geography



1. Layers of the Earth

Learning Guide

1. What is Geomorphology?

- **Geomorphology** = Study of the **changing surface of the Earth**
- The Earth's surface is shaped by:

Two Main Forces:

- **Internal forces (Endogenic):**
 - Come from **inside the Earth**
 - Example: tectonic movements, volcanoes, earthquakes
- **External forces (Exogenic):**
 - Act on the **surface of the Earth**
 - Example: weathering, erosion, deposition

Key Idea:

These forces work together to form **landforms** (mountains, valleys, plateaus)

2. Internal Structure of the Earth

- The Earth has **3 main layers**:
 1. **Crust**
 2. **Mantle**
 3. **Core**

3. The Crust (Outer Layer)

- **Outermost layer** where we live
- Thickness: **5 km – 75 km**
- Makes up only **1% of Earth's volume**

Composition:

- **Continental crust (SIAL):**
 - Made of **granite**
 - Minerals: Silicon + Aluminium
 - **Thicker (30–60 km)**
 - Less dense
- **Oceanic crust (SIMA):**
 - Made of **basalt**
 - Minerals: Silicon + Magnesium
 - **Thinner (5–15 km)**
 - More dense

Other Key Facts:

- Temperature: **-60°C to 200°C**
- Boundary with mantle: **Moho discontinuity**
- Crust **floats on the mantle**

Important: Continental crust = lighter, Oceanic crust = heavier

4. The Mantle (Middle Layer)

- Located **below the crust**
- Made of **peridotite rock**

- **Semi-fluid (plastic)** → not completely solid

Key Features:

- Temperature: **200°C – 3000°C**
- Thickness: **~1950 km**
- Makes up about **85% of Earth's volume**

Importance:

- Responsible for **movement of tectonic plates**
- Causes:
 - **Earthquakes**
 - **Volcanoes**

5. The Core (Innermost Layer)

- Made mainly of **iron and nickel**
- Divided into **two parts**:

Outer Core:

- **Liquid**
- Thickness: **~2200 km**
- Responsible for **Earth's magnetic field**

Inner Core:

- **Solid** (due to extreme pressure)
- Radius: **~1220 km**

 *Important:*

Even though it is very hot, the inner core is solid because of pressure

6. What is a Rock?

- A **rock** = solid made of **different minerals**
- Rocks are classified based on **how they form**

7. Types of Rocks

1. Igneous Rocks

- Form when **magma cools and solidifies**

Types:

- **Intrusive:**
 - Forms inside Earth
 - Example: granite
- **Extrusive:**
 - Forms on surface
 - Example: basalt

Characteristics:

- Hard and durable
- Often dark in colour
- Fine or coarse texture

Importance:

- Used in construction
- Contain minerals (gold, diamonds)
- Form **fertile soils when weathered**

2. Sedimentary Rocks

- Form from **deposited sediments**

Process:

 Transport → Deposition → Compaction → Cementation

Characteristics:

- Layered
- Often contain **fossils**
- Soft and easily eroded

Uses:

- Fertile soils
 - Contain coal, oil, gas
 - Used in cement (limestone)
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3. Metamorphic Rocks

- Form under **heat and pressure**
- Originate from existing rocks

Characteristics:

- Hardest rocks
- Oldest rocks
- Physically and chemically changed
- Often have **patterns/texture**

Uses:

- Building and decoration (e.g. marble)
 - Very durable
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8. The Rock Cycle

- Continuous process where rocks **change from one type to another**

Cycle:

- Magma → **Igneous rock**
- Weathering → **Sediment**
- Sediment → **Sedimentary rock**
- Heat & pressure → **Metamorphic rock**

 Important:

Rocks can change in **any order**, not just one direction

9. FINAL EXAM SUMMARY (VERY IMPORTANT)

- Earth has **3 layers: Crust, Mantle, Core**
 - Crust = thin, where life exists
 - Mantle = largest layer, drives plate movement
 - Core = hottest layer, produces magnetic field
 - Rocks are classified into:
 - **Igneous**
 - **Sedimentary**
 - **Metamorphic**
 - Rock cycle = **continuous change over time**
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